On Similarity Transformation Problems: Globally Optimal Results and Applications

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***Abstract*— Similarity transformation problems are important in robotic instrumentation and computer vision-based measurements since in many cases the information of visually observed scene scale is unknown and must be restored for accurate 3-D reconstruction. In existing solvers, the scale is often considered as a scalar, i.e., isotropic, which may be invalid for anisotropic-scale setups. This article exploits some mathematical coincidences that will lead to efficient solutions to these problems. Possible further applications also include hand-eye calibration and structure-from-motion. We revisit pose estimation problems within the framework of similarity transformation, the one that considers scale-stretching, rotation, and translation simultaneously. Two major problems are taken into account, i.e., the scale-stretching point-cloud registration (PCR) and perspective-n-points (PnPs). It has been found out that these two problems are quite similar. Moreover, we solve the anisotropic-scale registration problem which is important and is a remaining unsolved one in previous literatures. To compute the globally optimal solution of these nonconvex problems, the algebraic solution is obtained to compute all local minima using computationally efficient methods. The designed algorithm is deployed for robotic-arm pose estimation. We also extend the algorithm for solving the problem of robust magnetometer calibration. Visual pose experiments verify the superiority of the proposed method compared with representatives, including P3P, Lambda-Twist P3P, and EPnP, which can be reproduced by the repository in `https://github.com/zarathustr/APnP`.**

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NOMENCLATURE

$\|\cdot\|$	Euclidean norm.
$\boldsymbol{X}^\top$	Transpose of matrix $\boldsymbol{X}$.
$\boldsymbol{x}_\times$	Skew symmetric matrix of vector $\boldsymbol{x}$.
$\boldsymbol{q}$	Unit attitude quaternion.
$\boldsymbol{I}$	Identity matrix of proper size.
$\boldsymbol{0}$	Zeros matrix of proper size.
$\mathbb{R}^n$	Real n-dimensional vector space.
$\mathbb{R}^{n\times m}$	Real $n \times m$ matrix space.
$\mathbb{Z}$	Nonnegative integer space.
$\mathbb{R}^n_+$	Real n-dimensional vector space with positive entries.
$\mathfrak{N}(\boldsymbol{\gamma}, \boldsymbol{\Sigma})$	Normal distribution with mean of $\boldsymbol{\gamma}$ and covariance of $\boldsymbol{\Sigma}$.
$\mathrm{SO}(n)$	n-dimensional special orthogonal group.
$\mathrm{SE}(n)$	n-dimensional special Euclidean group.

I. INTRODUCTION

A. Motivation and Related Work

ACCURATE robotic navigation and mapping require precision pose estimation from visual measurements, including images and point-cloud information [1], [2]. Among well-developed methods, point-cloud registration (PCR) and perspective-n-points (PnPs) are two major categories for estimating relative or absolute poses. The PCR aims to determine the relative pose between successive 3-D/3-D point-cloud measurements. The PnP solves the camera pose by the correspondence between 2-D image points and 3-D world points. Registration-based methods can also be shifted to solving vision-based calibration problems [3], [4]. In these two methods, the determination of scale is vital, as it links the estimates to the geometry of the real world. Therefore, current research efforts are mainly devoted to obtaining accurate estimates of scale, rotation, and translation simultaneously.

Extensive efforts have been paid to solving PCR and PnP problems. Markley and Arun proposed solvers for point-to-point registration early back in the 1980s, respectively [5], [6]. The kernel problem of PCR is the determination of point

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correspondences between two point sets to be registered. This motivates the idea of iterative closest point (ICP) that was proposed in the 1990's [7], [8], [9]. The ICP is challenging, nonconvex, and NP-hard since the sizes of two point sets are not identical. In many registration problems, the scale factor is always unknown because of different measurement principles and unsatisfactory calibration. Horn solves this scale-stretching registration in [10] and [11] by analytical results. This was later introduced to the ICP for refined scale-factor determination [12]. For the PnP problem, the scale is the depth of the pixel that is significant to the size of 3-D reconstruction. Early methods use direct linear transform (DLT) [13] by vectorizing the rotation matrix. An approximate pose solution will be obtained and will be refined by taking the nearest rotation (orthonormalization) into account [14], [15]. The perspective-3-points (P3Ps) solves the PnP problem by taking three points of them and verifying the solution via an extern fourth point [16]. These methods are simple and computationally efficient but will suffer from large errors when the input measurements are noisy. The efficient PnP (EPnP) algorithm solves the method in a least-square manner with singular value decomposition [17]. There are other variants of PCR, e.g., the ones using various metrics of point-to-plane [18] and point-to-line [19] ones, which are not the objective of this article. Some other methods of PnP solve the problem by considering the scale as a nonlinear function of attitude and translation [20], [21]. The new problem is no longer a similarity transformation problem, which needs extensive efforts to solve the nonlinear optimization. We also do not discuss them in this article. In other words, this article targets at solving the problem analytically to a maximum extent. To this end, we must face the following challenges behind the related research.

1) *Challenge 1—Scale Matters:* We also notice that, following the research trend introduced above, there are still remaining problems in the field. First, the connection between these problems is not very clear. Furthermore, previous methods only consider the isotropic scale in estimation. However, in real-world scenarios, usually, practitioners will encounter anisotropic-scale PCR (APCR) problems. For instance, if we consider the registration of point sets measured by two different laser scanners, the scale may not be the same. Currently, there is a trend in PCR that merges the anisotropic concerns into the original one [22]. However, the anisotropic-scale registration problem is still not solved completely. It is currently challenging, as in the conclusion of [12], it has been pointed out that the anisotropic-scale registration is much harder than the isotropic one.
2) *Challenge 2—Convergence is Tough:* Although there have been further studies on the anisotropic registration by Du al. [23] and Li et al. [24], the results show that there exist local minima in their algorithm and satisfactory initial guess must be obtained to guarantee the global convergence, which may be false in terms of high outlier rate. This problem continuously exists in further related works [25], [26], which highly limits the practical performance.
3) *Challenge 3—More Applications Meet Registration:* In [27], it is inferred that the anisotropic-scale registration is highly related to sensor calibration issues, which are important in inertial navigation and robotics. But the problem is so nonconvex and only approximate optimization can be solved. Therefore, it will be a little bit difficult to transfer existing results of isotropic-scale registration to another. This article aims to solve these remaining issues that are important to the community. Theoretical and experimental results support our new findings. With contributions to be revealed in Section III, this article treats another challenge, i.e., the data association problem, as a well-solved one. The reason is that the data association problem, sometimes also referred to as the correspondence matching problem, is a completely different one, which receives popularity from diverse communities. In this way, this work will only use some mature data association techniques in the experimental validation stage.

B. Contributions

In this article, we revisit these problems by unifying them as the similarity transformation problem. The contributions of this article are summarized as follows.

1) First, through theoretical analysis, we show that scale-stretching PCR (SPCR) and PnP are very similar. In this way, a simple eigenvalue solution is designed that unifies them together.
2) We extend the isotropic-scale registration to the anisotropic-scale one. The new problem is nonconvex. Therefore, approximate and globally optimal solutions are derived to solve this challenging problem. The proposed globally optimal method is convergence-free while initial-value-free, leading to the fact that it is efficient and deterministic.
3) The developed theory is transferred to solve not only the anisotropic-scale registration and PnP but also it helps solve the problem of magnetometer calibration, where a point-to-surface registration scheme is proposed.

C. Outline

The remainder of the article is organized as follows: Sections II-B–II-D first introduce the details of several similarity transformation problems to be studied. Local and globally optimal solutions are then presented in Section III. Two experimental applications, results, and comparisons are shown in Section IV. Finally, concluding remarks are drawn in Section V.

II. Similarity Transformation Problems

A. Transformations

Some notations are defined in the Nomenclature at the beginning of this article. Rotation matrices distribute on the manifold of n-dimensional special orthogonal group such that

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$\mathrm{SO}(n) := \{\boldsymbol{R} \in \mathbb{R}^{n\times n} | \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}, \det(\boldsymbol{R}) = +1\}$. The special Euclidean group complements $\mathrm{SO}(n)$ with an additional translation vector

$$\mathrm{SE}(n) := \left\{ \begin{pmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0} & 1 \end{pmatrix} \middle| \boldsymbol{R} \in \mathrm{SO}(n), \boldsymbol{t} \in \mathbb{R}^n \right\} \tag{1}$$

in which $\boldsymbol{0}$ denotes a zero matrix with adequate size. When the scale factor is considered, the similarity transformation group is

$$\mathrm{Sim}(n) := \left\{ \begin{pmatrix} s\boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0} & 1 \end{pmatrix} \middle| s \in \mathbb{R}, \boldsymbol{R} \in \mathrm{SO}(n), \boldsymbol{t} \in \mathbb{R}^n \right\}.$$

In this article, $\mathrm{Sim}(n)$ is extended to $\mathrm{SIM}(n)$, that characterizes the scale factor in an anisotropic manner

$$\mathrm{SIM}(n) := \left\{ \begin{pmatrix} \boldsymbol{S}\boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0} & 1 \end{pmatrix} \middle| \boldsymbol{S} \in \mathrm{diag}\left(\mathbb{R}_+^n\right), \boldsymbol{R} \in \mathrm{SO}(n), \boldsymbol{t} \in \mathbb{R}^n \right\}$$

where $\mathbb{R}_+^n$ stands for the set of n-dimensional real positive vectors. For 3-D cases, rotation can be parameterized with unit quaternion, say $\boldsymbol{q} = (q_0, q_1, q_2, q_3)^\top$. Rotation matrix $\boldsymbol{R}$ is quadratically represented in terms of elements of $\boldsymbol{q}$, so the negative quaternion $-\boldsymbol{q}$ represents the same rotation as $\boldsymbol{R}$. Quaternion has the only constraint of unitary norm thus it is more convenient in optimization problems. Note that, quaternion is not the simplest formulation for 3-D rotation parameterization. For any n-dimensional rotation matrix $\boldsymbol{R}$, it has related Lie algebra $\boldsymbol{\xi}$ such that $\boldsymbol{R} = \exp(\boldsymbol{\xi}_\times)$, in which $\boldsymbol{\xi}_\times$ denotes a mapping from $\mathbb{R}^{n(n-1)/2}$ to $\mathbb{R}^{n\times n}$. However, there are infinity many $\boldsymbol{\xi}$ correspond to $\boldsymbol{R}$ since $\boldsymbol{\xi}$ is periodic. Thus, even for a 3-D case, the computation of Jacobian may be truncated using limited elements of expansion of $\exp(\boldsymbol{\xi}_\times)$. The introduction of Lie algebra simplifies the number of variables for identifying rotation but brings more nonlinearity. As a consequence, this article uses the unit quaternion as a kernel tool for the solution of optimization problems.

B. Problem I: Scale-Stretching 3-D Registration

What we are concerning the most in the remainder of this article includes the scale-SPCR and PnP problems. Given two 3-D point sets $\{\mathcal{B}\}$ and $\{\mathcal{R}\}$, the scale SPCR aligns the two sets together by solving the following absolute orientation problem:

$$\underset{s\in\mathbb{R}, \boldsymbol{R}\in\mathrm{SO}(3), \boldsymbol{t}\in\mathbb{R}^3}{\arg\min} \mathcal{L}_{\mathrm{SPCR}} = \sum_{i=1}^{\mathcal{N}} \|\boldsymbol{b}_i - s\boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t}\|^2 \tag{2}$$

in which $\boldsymbol{b}_i \in \{\mathcal{B}\}$ and $\boldsymbol{r}_i \in \{\mathcal{R}\}$ are points from two sets. Note that here we assume that the correspondences between $\{\mathcal{B}\}$ and $\{\mathcal{R}\}$ have been fixed. The scale- SPCR deals with the registration problem on Sim(3). The scale-stretching phenomenon usually comes from devices with different scale factors. Thus, it is frequently required for 3-D/3-D alignment between multiple-point cloud measurements from laser scanners.

C. Problem II: Perspective-n-Points

The PnP aims to solve the pose estimation problem between the undistorted 2-D points in the image plane and corresponding 3-D points in the world frame. Given image coordinates $\boldsymbol{u}_i \in \mathbb{R}^2$ for $i = 1, 2, \ldots, \mathcal{N}$ and their related 3-D world points $\boldsymbol{v} \in \mathbb{R}^3$ for $i = 1, 2, \ldots, \mathcal{N}$, one would like to achieve the following perspective transformation:

$$s\left(\boldsymbol{u}_i^\top, 1\right)^\top = \boldsymbol{K}(\boldsymbol{R}\boldsymbol{v}_i + \boldsymbol{t}) \tag{3}$$

where $\boldsymbol{K} \in \mathbb{R}^{3\times3}$ is an affine calibration matrix (intrinsic) accounting for the focal lengths and central points in horizontal and vertical directions, respectively. Denoting $\boldsymbol{b}_i = (\boldsymbol{u}_i^\top, 1)^\top$ and $\boldsymbol{r}_i = \boldsymbol{v}_i$, it is able for us to construct the optimization of

$$\underset{s\in\mathbb{R}, \boldsymbol{R}\in\mathrm{SO}(3), \boldsymbol{t}\in\mathbb{R}^3}{\arg\min} \mathcal{L}_{\mathrm{PnP}} = \sum_{i=1}^{\mathcal{N}} \|s\boldsymbol{b}_i - \boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t}\|^2. \tag{4}$$

Here, s, $\boldsymbol{R}$, and $\boldsymbol{t}$ form a transformation on Sim(3). Normally, s is not independent of $\boldsymbol{R}$, because from (3), one can conclude from the last line that s is in nonlinear form of $\boldsymbol{R}$ and $\boldsymbol{t}$. Therefore, the essential way for solving highly accurate PnP relies on the optimization of bundle adjustment (BA). However, to obtain a good initial guess of BA, s can be treated independently of $\boldsymbol{R}$ and $\boldsymbol{t}$, such is a common practice in popular solvers like EPnP [17].

D. Problem III: Anisotropic-Scale Registration

In registration with anisotropic scale factors, the following nonlinear least square problem is considered:

$$\underset{\boldsymbol{R}\in\mathrm{SO}(3), \boldsymbol{t}\in\mathbb{R}^3, \boldsymbol{S}\in\mathrm{diag}\left(\mathbb{R}_+^3\right)}{\arg\min} \mathcal{L}_{\mathrm{APCR}} = \sum_{i=1}^{\mathcal{N}} \|\boldsymbol{b}_i - \boldsymbol{S}\boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t}\|^2 \tag{5}$$

in which $\boldsymbol{b}_i$ and $\boldsymbol{r}_i$ are ith correspondence point pair from two point sets $\{\mathcal{B}\}$ and $\{\mathcal{R}\}$, respectively. $\boldsymbol{R}$ denotes the rotation matrix that distributes on the special orthogonal group $SO(3)$ subject to the nonlinear constraints $\boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{I}, \det(\boldsymbol{R}) = 1$. $\boldsymbol{t}$ acts as a translation vector while $\boldsymbol{S}$ contains three positive anisotropic scale factors such that $\boldsymbol{S} = \mathrm{diag}(s_1, s_2, s_3)$ with s_1, s_2, and s_3 being positive numbers in the real positive set $\mathbb{R}_+$. Here, $\boldsymbol{S}$, $\boldsymbol{R}$, and $\boldsymbol{t}$ constitute a transformation on SIM(3). Previously, in [12], a scale-stretching ICP problem has been studied where $\boldsymbol{S}$ degenerates to a scalar, which is exactly the scale-stretching registration problem stated above. The APCR problem is much more challenging than the previous one as mentioned in the concluding remarks of [12], where strong coupling between $\boldsymbol{S}$ and $\boldsymbol{R}$ has been illustrated. Therefore, the problem is challenging in the aspect of a globally optimal optimization solution.

III. Solutions

A. Scale-Stretching Registration and PnP

The elements of $\boldsymbol{b}_i$ and $\boldsymbol{r}_i$ are $\boldsymbol{b}_i = (b_{i,1}, b_{i,2}, b_{i,3})^\top$ and $\boldsymbol{r}_i = (r_{i,1}, r_{i,2}, r_{i,3})^\top$, respectively. Equations (2) and (4) specify the target objective of the scale-SPCR and PnP. From these expressions, it is clear that the only difference is the location of s. In the following contents, we are going to show some algebraic results for solving these problems. Moreover, the two problems are eventually solved in a unified fashion.

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The objective of PnP can be extended as

$$\mathcal{L}_{\text{PnP}} = \sum_{i=1}^{\mathcal{N}} (s\boldsymbol{b}_i - \boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t})^\top (s\boldsymbol{b}_i - \boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t}). \tag{6}$$

Introducing the unit quaternion $\boldsymbol{q}$ for attitude parameterization of $\boldsymbol{R}$, the Lagrangian is

$$\tilde{\mathcal{L}}_{\text{PnP}} = \mathcal{L}_{\text{PnP}} + \lambda(\boldsymbol{q}^\top \boldsymbol{q} - 1) \tag{7}$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier. Here, a detailed relationship between a unit quaternion and a rotation matrix can be found in [28]. Then, all local optimum occur at the place where the Jacobian is zero. To compute the Jacobian analytically, some previous results are invoked. The rotation matrix $\boldsymbol{R} = (\boldsymbol{P}_1\boldsymbol{q}, \boldsymbol{P}_2\boldsymbol{q}, \boldsymbol{P}_3\boldsymbol{q})$ in which $\boldsymbol{P}_1$, $\boldsymbol{P}_2$ and $\boldsymbol{P}_3$ are linear matrix of $\boldsymbol{q}$ [29], [30], thus

$$\boldsymbol{R}\boldsymbol{r}_i = (\boldsymbol{P}_1\boldsymbol{q}, \boldsymbol{P}_2\boldsymbol{q}, \boldsymbol{P}_3\boldsymbol{q})\boldsymbol{r}_i = \sum_{j=1}^{3} r_{i,j}\boldsymbol{P}_j\boldsymbol{q} \tag{8}$$

which gives

$$\partial \boldsymbol{b}_i^\top \boldsymbol{R}\boldsymbol{r}_i/\partial \boldsymbol{q} = \sum_{j=1}^{3} r_{i,j}\boldsymbol{P}_j^\top \boldsymbol{b}_i = \sum_{j=1}^{3} r_{i,j}\boldsymbol{M}_j(\boldsymbol{b}_i)\boldsymbol{q} \tag{9}$$

where $\boldsymbol{M}_j$ matrix is linear in the form of $\boldsymbol{b}_i$ (see [30]). In this way, the blocks of Jacobian are

$$\frac{\partial \tilde{\mathcal{L}}_{\text{PnP}}}{\partial \boldsymbol{q}} = 2\lambda\boldsymbol{q} - 2\sum_{i=1}^{\mathcal{N}}\sum_{j=1}^{3} r_{i,j}\boldsymbol{M}_j(s\boldsymbol{b}_i - \boldsymbol{t})\boldsymbol{q} \tag{10}$$

$$\frac{\partial \tilde{\mathcal{L}}_{\text{PnP}}}{\partial \boldsymbol{t}} = 2\mathcal{N}\boldsymbol{t} - 2\sum_{i=1}^{\mathcal{N}} s\boldsymbol{b}_i - \boldsymbol{R}\boldsymbol{r}_i \tag{11}$$

$$\frac{\partial \tilde{\mathcal{L}}_{\text{PnP}}}{\partial s} = 2\sum_{i=1}^{\mathcal{N}} \boldsymbol{b}_i^\top (s\boldsymbol{b}_i - \boldsymbol{R}\boldsymbol{r}_i - \boldsymbol{t}). \tag{12}$$

The optimality meets $\nabla\tilde{\mathcal{L}}_{\text{PnP}} = \boldsymbol{0}$. Then, one has

$$\boldsymbol{t} = s\bar{\boldsymbol{b}} - \boldsymbol{R}\bar{\boldsymbol{r}} \tag{13}$$

$$\lambda\boldsymbol{q} = \sum_{i=1}^{\mathcal{N}}\sum_{j=1}^{3} r_{i,j}\boldsymbol{M}_j\big[s\big(\boldsymbol{b}_i - \bar{\boldsymbol{b}}\big) + \boldsymbol{R}\bar{\boldsymbol{r}}\big]\boldsymbol{q} \tag{14}$$

$$s = \left[\sum_{i=1}^{\mathcal{N}} \boldsymbol{b}_i^\top \boldsymbol{R}\big(\boldsymbol{r}_i - \bar{\boldsymbol{r}}\big)\right] \Big/ \left[\sum_{i=1}^{\mathcal{N}} \boldsymbol{b}_i^\top \big(\boldsymbol{b}_i - \bar{\boldsymbol{b}}\big)\right]. \tag{15}$$

Introducing the identities

$$\boldsymbol{P}_i^\top \boldsymbol{P}_i\boldsymbol{q} = \boldsymbol{q} \tag{16}$$

$$\big(\boldsymbol{P}_i^\top \boldsymbol{P}_j + \boldsymbol{P}_j^\top \boldsymbol{P}_i\big)\boldsymbol{q} = \boldsymbol{0} \tag{17}$$

the term $\boldsymbol{M}_j(\boldsymbol{R}\bar{\boldsymbol{r}})\boldsymbol{q}$ can be simplified to scaled form of $\boldsymbol{q}$, say $\alpha\boldsymbol{q}$. Therefore, $\boldsymbol{q}$ will be solved via the following problem:

$$s\boldsymbol{W}\boldsymbol{q} = (\lambda - \alpha)\boldsymbol{q} \tag{18}$$

in which

$$\boldsymbol{W} = 1/\mathcal{N} \sum_{i=1}^{\mathcal{N}}\sum_{j=1}^{3} r_{i,j}\boldsymbol{M}_j\big(\boldsymbol{b}_i - \bar{\boldsymbol{b}}\big). \tag{19}$$

Note that, for this problem, $\boldsymbol{q}$ is an eigenvector of $\boldsymbol{W}$ associated with the eigenvalue of $(\lambda - \alpha)/s$. $\boldsymbol{W}$ is a 4×4 symmetric matrix so there will be four real eigenvalues corresponding to four possible local minima. Inserting these four possible solutions of $\boldsymbol{q}$ back into (13) gives four possible solutions of s. The optimal set of solutions is then obtained by checking the loss-function value $\mathcal{L}_{\text{PnP}}$. These techniques are similar to the SPCR problem. The Lagrangian is

$$\tilde{\mathcal{L}}_{\text{SPCR}} = \mathcal{L}_{\text{SPCR}} + \lambda\big(\boldsymbol{q}^\top \boldsymbol{q} - 1\big). \tag{20}$$

We use the fact that $\boldsymbol{R}$ can be decomposed into

$$\boldsymbol{R}^\top = \big(\boldsymbol{Q}_1\boldsymbol{q}, \boldsymbol{Q}_2\boldsymbol{q}, \boldsymbol{Q}_3\boldsymbol{q}\big) \tag{21}$$

in which $\boldsymbol{Q}_1$, $\boldsymbol{Q}_2$, and $\boldsymbol{Q}_3$ are linear matrices of $\boldsymbol{q}$. The derivative of the loss function with respect to quaternion is

$$\frac{\partial \tilde{\mathcal{L}}_{\text{SPCR}}}{\partial \boldsymbol{q}} = 2\lambda\boldsymbol{q} - \left(\sum_{i=1}^{\mathcal{N}}\sum_{j=1}^{3} s r_{i,j}\boldsymbol{M}_j\big[\big(\boldsymbol{b}_i - \bar{\boldsymbol{b}}\big)\big] + r_{i,j}\boldsymbol{I}\right)\boldsymbol{q} \tag{22}$$

indicating that the solution is also an eigenvalue problem, like (18).

B. Anisotropic-Scale Registration

For the anisotropic registration, from single-pair equation $\boldsymbol{b}_i = \boldsymbol{S}\boldsymbol{R}\boldsymbol{r}_i + \boldsymbol{t}$, one can conclude the averaged form as $\bar{\boldsymbol{b}} = \boldsymbol{S}\boldsymbol{R}\bar{\boldsymbol{r}} + \boldsymbol{t}$, that further gives a new optimization

$$\underset{\boldsymbol{R}\in\text{SO}(3),\boldsymbol{S}\in\text{diag}(\mathbb{R}_+^3)}{\arg\min}\ \mathcal{L}_{\text{APCR},1} = \sum_{i=1}^{\mathcal{N}} \big\|\tilde{\boldsymbol{b}}_i - \boldsymbol{S}\boldsymbol{R}\tilde{\boldsymbol{r}}_i\big\|^2 \tag{23}$$

which is decentralized one without translation $\boldsymbol{t}$ with $\tilde{\boldsymbol{b}}_i = \boldsymbol{b}_i - \bar{\boldsymbol{b}} = (b_{i,1}, b_{i,2}, b_{i,3})^\top$, $\tilde{\boldsymbol{r}}_i = \boldsymbol{r}_i - \bar{\boldsymbol{r}} = (r_{i,1}, r_{i,2}, r_{i,3})^\top$. Then, the following norm-based optimization is constructed:

$$\underset{\boldsymbol{S}\in\text{diag}(\mathbb{R}_+^3)}{\arg\min}\ \mathcal{L}_{\text{APCR},\boldsymbol{S}} = \sum_{i=1}^{\mathcal{N}} \big(\tilde{\boldsymbol{b}}_i^\top \boldsymbol{S}^{-2}\tilde{\boldsymbol{b}}_i - \tilde{\boldsymbol{r}}_i^\top \tilde{\boldsymbol{r}}_i\big)^2. \tag{24}$$

The Jacobian of $\mathcal{L}_{\text{APCR},\boldsymbol{S}}$ is

$$\frac{\partial \mathcal{L}_{\text{APCR},\boldsymbol{S}}}{\partial \boldsymbol{s}} = 4\sum_{i=1}^{\mathcal{N}} \begin{pmatrix} b_{i,1}^2\beta/s_1^3 \\ b_{i,2}^2\beta/s_2^3 \\ b_{i,3}^2\beta/s_3^3 \end{pmatrix} \tag{25}$$

in which $\boldsymbol{s} = (s_1, s_2, s_3)^\top$ and $\beta = (\|\tilde{\boldsymbol{r}}_i\|^2 - b_{i,1}^2/s_1^2 - b_{i,2}^2/s_2^2 - b_{i,3}^2/s_3^2)$. To solve $\boldsymbol{s}$, the equation $\partial\mathcal{L}_{\text{APCR},\boldsymbol{S}}/\partial\boldsymbol{s} = \boldsymbol{0}$ must be solved. This results in the following polynomial equation after simplification:

$$\begin{cases} g_2g_3a_{11} + g_1g_3a_{12} + g_1g_2a_{13} + g_1g_2g_3a_{14} = 0 \\ g_2g_3a_{21} + g_1g_3a_{22} + g_1g_2a_{23} + g_1g_2g_3a_{24} = 0 \\ g_2g_3a_{31} + g_1g_3a_{32} + g_1g_2a_{33} + g_1g_2g_3a_{34} = 0 \end{cases} \tag{26}$$

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with $g_1 = s_1^2$, $g_2 = s_2^2$, $g_3 = s_3^2$, and

$$\begin{aligned}
&a_{11} = -\sum b_{i,1}^4, \quad a_{12} = -\sum b_{i,1}^2 b_{i,2}^2, \quad a_{13} = -\sum b_{i,1}^2 b_{i,3}^2 \\
&a_{14} = \sum b_{i,1}^2 \left\|\tilde{\boldsymbol{r}}_i\right\|^2, \quad a_{21} = -\sum b_{i,1}^2 b_{i,2}^2, \quad a_{22} = -\sum b_{i,2}^4 \\
&a_{23} = -\sum b_{i,2}^2 b_{i,3}^2, \quad a_{24} = \sum b_{i,2}^2 \left\|\tilde{\boldsymbol{r}}_i\right\|^2 \\
&a_{31} = -\sum b_{i,1}^2 b_{i,3}^2, \quad a_{32} = -\sum b_{i,2}^2 b_{i,3}^2 \\
&a_{33} = -\sum b_{i,3}^4, \quad a_{34} = \sum b_{i,3}^2 \left\|\tilde{\boldsymbol{r}}_i\right\|^2.
\end{aligned} \tag{27}$$

The closed-form solution to g_1, g_2, and g_3 is

$$g_1 = -G/G_1, \quad g_2 = G/G_2, \quad g_3 = -G/G_3 \tag{28}$$

where $G = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$, $G_1 = a_{12}a_{23}a_{34} - a_{12}a_{24}a_{33} - a_{13}a_{22}a_{34} + a_{13}a_{24}a_{32} + a_{14}a_{22}a_{33} - a_{14}a_{23}a_{32}$, $G_2 = a_{11}a_{23}a_{34} - a_{11}a_{24}a_{33} - a_{13}a_{21}a_{34} + a_{13}a_{24}a_{31} + a_{14}a_{21}a_{33} - a_{14}a_{23}a_{31}$ and $G_3 = a_{11}a_{22}a_{34} - a_{11}a_{24}a_{32} - a_{12}a_{21}a_{34} + a_{12}a_{24}a_{31} + a_{14}a_{21}a_{32} - a_{14}a_{22}a_{31}$. Then, $\boldsymbol{s}$ can be obtained via

$$s_1 = \sqrt{g_1}, \quad s_2 = \sqrt{g_2}, \quad s_3 = \sqrt{g_3}. \tag{29}$$

It should be noticed that (24) only considers the norm constraint so the coupling between $\boldsymbol{S}$ and $\boldsymbol{R}$ is lost. Thus, (29) is an approximate solution of $\boldsymbol{s}$.

Once an approximate set of scale factors has been found, inserting it back into (23) gives an approximate rotation estimate, say $\boldsymbol{R}_0$. This step can be achieved via either SVD [6] or eigen-decomposition (EIG) [7]. Using the approximate scale $\boldsymbol{S}_0$, the following equality for ith vector pair can be achieved:

$$\tilde{\boldsymbol{b}}_i = \boldsymbol{S}_0\tilde{\boldsymbol{S}}\tilde{\boldsymbol{R}}\boldsymbol{R}_0\tilde{\boldsymbol{r}}_i \tag{30}$$

where $\boldsymbol{S}_0\tilde{\boldsymbol{S}} = \boldsymbol{S}$ and $\tilde{\boldsymbol{R}}\boldsymbol{R}_0 = \boldsymbol{R}$ are true values of scale and rotation. $\tilde{\boldsymbol{S}}$ and $\tilde{\boldsymbol{R}}$ are error states of scale and rotation, respectively. By setting $\boldsymbol{u}_i = \boldsymbol{S}_0^{-1}\tilde{\boldsymbol{b}}_i$ and $\boldsymbol{v}_i = \boldsymbol{R}_0\tilde{\boldsymbol{r}}_i$, a new equivalent optimization for the error states can be established

$$\underset{\tilde{\boldsymbol{R}}\in\mathrm{SO}(3),\tilde{\boldsymbol{S}}\in\mathrm{diag}(\mathbb{R}_+^3)}{\arg\min} \sum_{i=1}^{\mathcal{N}} \left\|\boldsymbol{u}_i - \tilde{\boldsymbol{S}}\tilde{\boldsymbol{R}}\boldsymbol{v}_i\right\|^2 \tag{31}$$

which can be solved recursively via the formulae presented above. By repeating such construction over and over again, we gradually achieve the minimum. Note that $\tilde{\boldsymbol{R}}$ will eventually converge to identity matrix $\boldsymbol{I}$ but the convergence rate will be quite slow for final iterations. To solve this problem efficiently, we consider $\tilde{\boldsymbol{R}}$ to be a small-angle rotation, which can be parameterized as $\boldsymbol{R} = \boldsymbol{I} + \boldsymbol{\theta}_\times$, in which $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)^\top$ characterize small Euler angles and $\boldsymbol{\theta}_\times$ is its associated skew-symmetric matric. Denoting $\boldsymbol{x} = (\boldsymbol{\theta}^\top, \boldsymbol{s}^\top)^\top$, solving $\boldsymbol{x}$ optimally is identical to find all roots by zeroing the Lagrangian derivative. Simplified polynomial system is given by (32), shown at the bottom of the page, where h_{ij} are coefficient for the ith equation's jth monomial. Note that in this system, many coefficients can be omitted by reducing variables via the following equalities:

$$\begin{aligned}
&h_{12} = -h_{11}, \quad h_{21} = -h_{18}, \quad h_{22} = h_{18}, \quad h_{23} = h_{13} \\
&h_{27} = h_{17}, \quad h_{28} = h_{11}, \quad h_{31} = h_{17}, \quad h_{32} = -h_{17} \\
&h_{33} = h_{14}, \quad h_{34} = h_{24}, \quad h_{37} = h_{18}, \quad h_{38} = h_{11} \\
&h_{42} = h_{24}, \quad h_{43} = h_{13}, \quad h_{44} = h_{14}, \quad h_{45} = h_{25} \\
&h_{46} = h_{35}, \quad h_{47} = -2\,h_{18}, \quad h_{48} = 2\,h_{17}, \quad h_{49} = 2\,h_{11} \\
&h_{52} = h_{14}, \quad h_{53} = h_{13}, \quad h_{54} = h_{24}, \quad h_{55} = h_{15} \\
&h_{56} = h_{36}, \quad h_{57} = 2\,h_{11}, \quad h_{58} = -2\,h_{17}, \quad h_{59} = 2\,h_{18} \\
&h_{62} = h_{13}, \quad h_{63} = h_{14}, \quad h_{64} = h_{24}, \quad h_{65} = h_{16} \\
&h_{66} = h_{26}, \quad h_{67} = -2\,h_{11}, \quad h_{68} = 2\,h_{18}
\end{aligned} \tag{33}$$

and $h_{69} = 2\,h_{17}$. Required coefficients are summarized in the following equation:

$$\begin{aligned}
&h_{11} = \sum_{i=1}^{\mathcal{N}} v_{i,2}v_{i,3}, \quad h_{13} = \sum_{i=1}^{\mathcal{N}} v_{i,3}^2, \quad h_{14} = \sum_{i=1}^{\mathcal{N}} v_{i,2}^2 \\
&h_{15} = \sum_{i=1}^{\mathcal{N}} u_{i,2}v_{i,3}, \quad h_{16} = -\sum_{i=1}^{\mathcal{N}} u_{i,3}v_{i,2} \\
&h_{17} = -\sum_{i=1}^{\mathcal{N}} v_{i,1}v_{i,2}, \quad h_{18} = -\sum_{i=1}^{\mathcal{N}} v_{i,1}v_{i,3}, \quad h_{24} = \sum_{i=1}^{\mathcal{N}} v_{i,1}^2 \\
&h_{25} = -\sum_{i=1}^{\mathcal{N}} u_{i,1}v_{i,3}, \quad h_{26} = \sum_{i=1}^{\mathcal{N}} u_{i,3}v_{i,1}, \quad h_{35} = \sum_{i=1}^{\mathcal{N}} u_{i,1}v_{i,2} \\
&h_{36} = -\sum_{i=1}^{\mathcal{N}} u_{i,2}v_{i,1}, \quad h_{41} = -\sum_{i=1}^{\mathcal{N}} u_{i,1}v_{i,1}
\end{aligned} \tag{34}$$

and $h_{51} = -\Sigma_{i=1}^{\mathcal{N}} u_{i,2}v_{i,2}$, $h_{61} = -\Sigma_{i=1}^{\mathcal{N}} u_{i,3}v_{i,3}$. It should be noticed that from last three subequations of (32), one can directly solve s_1, s_2, and s_3 in terms of θ_1, θ_2, and θ_3, i.e.,

$$\begin{aligned}
s_1 &= -(h_{41} + h_{45}\theta_2 + h_{46}\theta_3)/ \\
&\quad \left(h_{43}\theta_2^2 + h_{49}\theta_2\theta_3 + h_{47}\theta_2 + h_{44}\theta_3^2 + h_{48}\theta_3 + h_{42}\right) \\
s_2 &= -(h_{51} + h_{55}\theta_1 + h_{56}\theta_3)/ \\
&\quad \left(h_{53}\theta_1^2 + h_{59}\theta_1\theta_3 + h_{57}\theta_1 + h_{54}\theta_3^2 + h_{58}\theta_3 + h_{52}\right) \\
s_3 &= -(h_{61} + h_{65}\theta_1 + h_{66}\theta_2)/ \\
&\quad \left(h_{63}\theta_1^2 + h_{69}\theta_1\theta_2 + h_{67}\theta_1 + h_{64}\theta_2^2 + h_{68}\theta_2 + h_{62}\right).
\end{aligned} \tag{35}$$

$$\begin{cases}
h_{15}s_2 + h_{16}s_3 + h_{11}s_2^2 + h_{12}s_3^2 + h_{13}s_2^2\theta_1 + h_{14}s_3^2\theta_1 + h_{17}s_3^2\theta_2 + h_{18}s_2^2\theta_3 = 0 \\
h_{25}s_1 + h_{26}s_3 + h_{21}s_1^2 + h_{22}s_3^2 + h_{23}s_1^2\theta_2 + h_{24}s_3^2\theta_2 + h_{27}s_3^2\theta_1 + h_{28}s_1^2\theta_3 = 0 \\
h_{35}s_1 + h_{36}s_2 + h_{31}s_1^2 + h_{32}s_2^2 + h_{33}s_1^2\theta_3 + h_{34}s_2^2\theta_3 + h_{37}s_2^2\theta_1 + h_{38}s_1^2\theta_2 = 0 \\
h_{41} + h_{42}s_1 + h_{45}\theta_2 + h_{46}\theta_3 + h_{43}s_1\theta_2^2 + h_{44}s_1\theta_3^2 + h_{47}s_1\theta_2 + h_{48}s_1\theta_3 + h_{49}s_1\theta_2\theta_3 = 0 \\
h_{51} + h_{52}s_2 + h_{55}\theta_1 + h_{56}\theta_3 + h_{53}s_2\theta_1^2 + h_{54}s_2\theta_3^2 + h_{57}s_2\theta_1 + h_{58}s_2\theta_3 + h_{59}s_2\theta_1\theta_3 = 0 \\
h_{61} + h_{62}s_3 + h_{65}\theta_1 + h_{66}\theta_2 + h_{63}s_3\theta_1^2 + h_{64}s_3\theta_2^2 + h_{67}s_3\theta_1 + h_{68}s_3\theta_2 + h_{69}s_3\theta_1\theta_2 = 0
\end{cases} \tag{32}$$

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Thus replacing s_1, s_2, and s_3 with (35) produces a system with unknowns of θ_1, θ_2, and θ_3 only. The transformed system is a little bit sophisticated. Related monomials are (36), as shown at the bottom of the next page. To solve this nonlinear polynomial system, we may use the Gröbner-basis method. Gröbner bases are common properties of a certain polynomial system. They actually represent a reduced form of original polynomials so they are easier to be solved via variable elimination. However, the Gröbner-basis method is not applicable for solving this system since the reduced Gröbner bases are too complicated so that evaluating these bases consumes much more time even than numerical optimizers. Therefore, to efficiently solve the new system containing only elements of $\boldsymbol{\theta}$, a linear equation is appended: $H = 1 + \theta_1 + \theta_2 + \theta_3$, which is a linear combination of first-order monomials $\{1, \theta_1, \theta_2, \theta_3\}$. Since the magnitude of $\boldsymbol{\theta}$ is small, high-order terms will be very tiny. In this way, we depart $\boldsymbol{\zeta}_{\boldsymbol{\theta}}$ into two subgroups $\boldsymbol{\zeta}_1$ and $\boldsymbol{\xi}_2$, where $\boldsymbol{\zeta}_1$ contains all terms with orders of no more than 3 and $\boldsymbol{\zeta}_2$ consists of the remainder. Stacking all coefficient of original polynomials of $\boldsymbol{\theta}$ forms the Macaulay matrix $\mathcal{M}$ such that

$$\mathcal{M}\boldsymbol{\zeta}_{\boldsymbol{\theta}} = \mathbf{0}. \tag{37}$$

However, after extending H to the original system, the H is not zero in general. Denoting the new Macaulay matrix as $\bar{\mathcal{M}}$, using H as the first extended equation of the new polynomial, we can write $\bar{\mathcal{M}}$ as

$$\bar{\mathcal{M}} = \begin{pmatrix} \bar{\mathcal{M}}_1 & \bar{\mathcal{M}}_2 \\ \bar{\mathcal{M}}_3 & \bar{\mathcal{M}}_4 \end{pmatrix} \tag{38}$$

so that

$$\left(H\boldsymbol{\zeta}_1^\top, \mathbf{0}\right)^\top = \bar{\mathcal{M}}\left(\boldsymbol{\zeta}_1^\top, \boldsymbol{\zeta}_2^\top\right)^\top. \tag{39}$$

Using the identity of Schur complement, we have

$$\mathcal{K} = \bar{\mathcal{M}}_1 - \bar{\mathcal{M}}_2\bar{\mathcal{M}}_4^{-1}\bar{\mathcal{M}}_3. \tag{40}$$

So it follows that:

$$\mathcal{F}\boldsymbol{\zeta}_1 = H\boldsymbol{\zeta}_1 \tag{41}$$

where $\boldsymbol{\zeta}_1$ is the eigenvector of $\mathcal{F}$ associated with eigenvalue H. In this way, all 27 eigenvectors of $\mathcal{F}$ can be obtained. Not all of these 27 eigenvectors are real. Then, by selecting the real vectors and inserting them back to the loss function value, the global optimal solution will be obtained. Since the EIG is highly numerical stable, the proposed method is also numerical stable.

IV. Experimental Results

In this section, the experimental results from various sensors are reported. The proposed method has been applied to these cases where comparisons with representatives are systematically conducted. In all these experiments, the data association has been performed using a k-d tree, which is easy to implement and classical.

TABLE I
Success Rates of Different Anisotropic Registration Methods

Initial	Du et al. [23]	Li et al. [24]	Chen et al. [25]	Proposed Global
Random	14.686%	18.922%	16.815%	100%
Author-Given	74.238%	63.401%	78.736%	100%

A. Synthetic Evaluation: Comparisons With Existing Anisotropic Registration Methods

We replicate the anisotropic registration algorithms in [23], [24], and [25]. These three methods all belong to the iterative algorithms which need a good initial guess to converge to the global minimum. We simulate randomly sampled point cloud data from the Stanford Bunny model [31]. We sample 50 cases with 5000 points to conduct registration. For each case, first, 100 initial guess values are generated randomly for convergence analysis. For reference and transformed point pairs, the point numbers are consistent, i.e., there will be no data association challenges in this test. Then, we conduct another test using initial guess values provided in the original works (marked as author-given). The registration success rates are averaged and summarized in Table I. Seen from the results, since the proposed method is globally optimal, the method is free of initial-value selection and iterations. However, other representatives are all iterative. Thus, the performance is largely dependent on the quality of the initial guess. Although all these methods provide their respective initial value computing strategy in their works, they do not always lead to satisfactory registration results. This shows that our method is much more reliable and deterministic compared to these existing ones. We also show anisotropic registration results of several open models. We utilize the `frog`[1] model and the `armadillo`[2] model for validation, whose standard models are shown in Fig. 1. The standard models are downsampled to high-resolution point clouds. Gaussian noises with covariance of $\boldsymbol{\Sigma} = 10^{-2}$ m^2 are added to the models, denoting a typical noisy sensor specification for modern 3-D LiDARs. The scale factors for two different models are emulated as 2.5 and 0.45, respectively. In these tests, rotation matrices, and translation vectors are generated randomly as ground truth. The evaluated registration results with estimated correspondences are shown in Figs. 2 and 3.

Some representative candidates are employed for comparisons. We show the registration accuracy results for `frog` model in Figs. 4 and 5, with statistics shown in Table II. The error metrics are

$$\begin{aligned} \mathcal{E}_{\text{rot}} &= \arccos\frac{\text{tr}\left(\boldsymbol{R}^\top\boldsymbol{R}_{\text{true}}\right) - 1}{2} \\ \mathcal{E}_{\text{trans}} &= \|\boldsymbol{t} - \boldsymbol{t}_{\text{true}}\|. \end{aligned} \tag{42}$$

The results indicate that the proposed method achieves the best rotation and translation accuracy in these tests. As shown in the correspondences estimated previously, we may see that the proposed method can generate highly accurate point correspondences given noisy point cloud pairs. This shows

[1] http://visionair.ge.imati.cnr.it/ontologies/shapes
[2] https://graphics.stanford.edu/data/3Dscanrep

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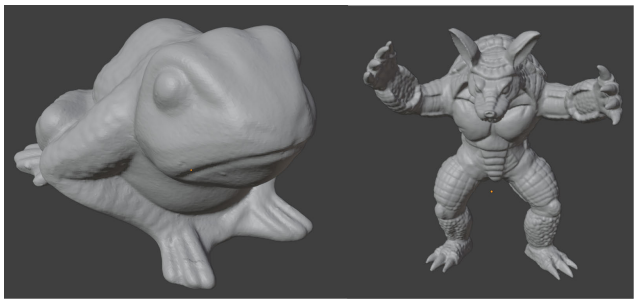


Fig. 1. Utilized standard models for accuracy comparisons of 3-D anisotropic registration methods.

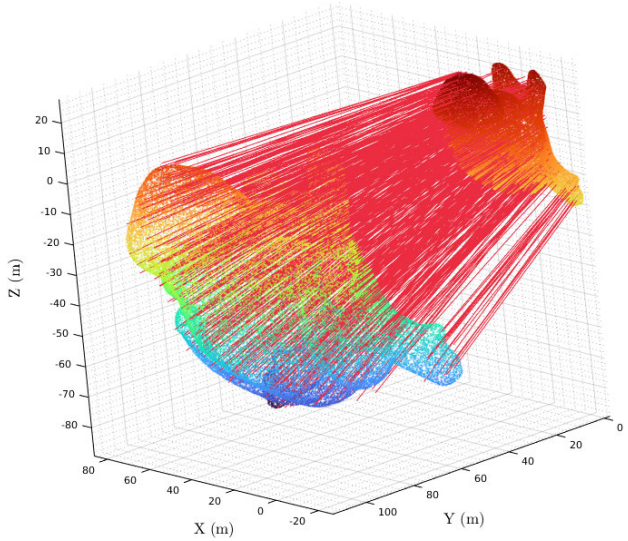


Fig. 2. Registration results with the proposed anisotropic scale estimation of the noisy `frog` models with estimated inlier correspondences (marked in red).

that the method would be capable of estimating accurate poses encountering data with outliers.

B. Application I: Accurate PnP Camera Pose

When conducting dynamic grasping tasks with a dynamic camera, the camera needs to understand the pose of the robotic gripper in real-time. This requires accurate perception of the gripper pose by visual correspondences. In our platform shown in Fig. 6, there is a gripper installed on a 6-degree-of-freedom (6-DOF) robotic manipulator. There is a 4×4 mini chessboard pattern attached to the robotic gripper using standard printing.

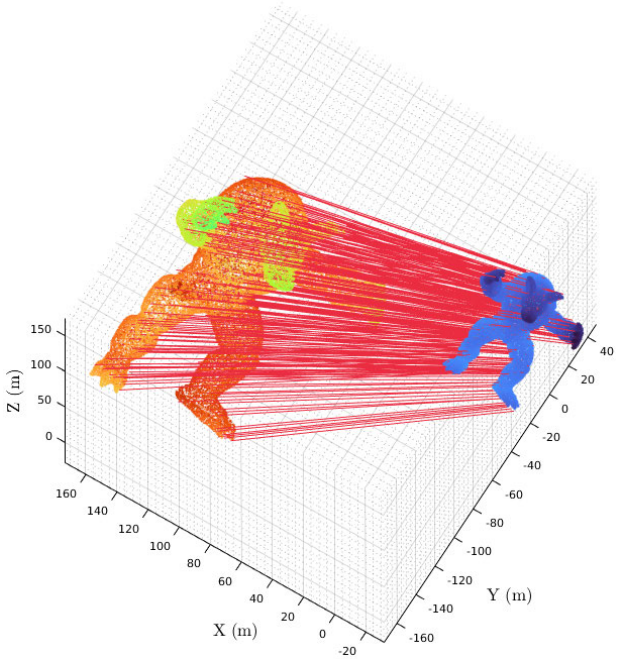


Fig. 3. Registration results with the proposed anisotropic scale estimation of the noisy `armadillo` models with estimated inlier correspondences (marked in red).

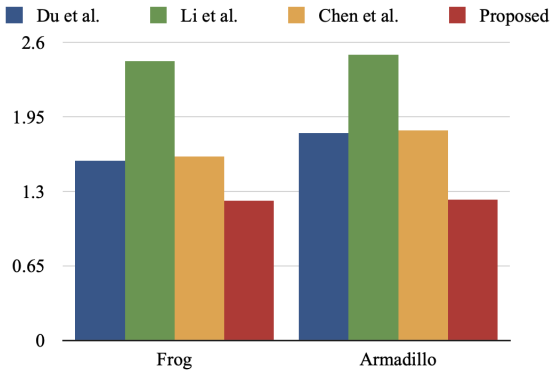


Fig. 4. Rotation errors of the proposed method with representative methods. The unit is degree.

A dynamically moving ZED-M stereo camera gazes at the pattern on the gripper so that the motion of the gripper can be inspected. To compute the relative camera pose (left camera) with respect to this mini gripper pattern, we implement

$$\boldsymbol{\zeta}_{\boldsymbol{\theta}}^{\top} = \left\{\theta_1^{\kappa_1}\theta_2^{\kappa_2}\theta_3^{\kappa_3} | \kappa_1, \kappa_2, \kappa_3 \in \mathbb{Z}_+\right\}$$
$$= \left(\begin{array}{l} 1, \theta_1, \theta_1^4, \theta_1^5, \theta_1^2, \theta_1^3, \theta_1^6, \theta_2^2, \theta_2^3, \theta_3^2, \theta_3^3, \theta_2^4, \theta_3^4, \theta_1\theta_2^2, \theta_1\theta_3^2, \theta_2^2\theta_3, \theta_2\theta_3^2, \theta_1^2\theta_2^2, \theta_2^3\theta_3^2, \theta_1^2\theta_3^2, \theta_2^2\theta_3^3, \theta_3, \theta_2, \theta_1^3\theta_2^2\theta_3, \theta_1^2\theta_2^2\theta_3, \\ \theta_1^2\theta_2\theta_3^2, \theta_1^3\theta_2\theta_3^2, \theta_1\theta_2^4\theta_3^2, \theta_1\theta_2^2\theta_3^4, \theta_1\theta_3, \theta_1\theta_2, \theta_1^2\theta_2, \theta_1^4\theta_2, \theta_1^2\theta_3, \theta_1^4\theta_3, \theta_2\theta_3, \theta_1^3\theta_3, \theta_1^3\theta_2, \theta_1^6\theta_2, \theta_1^5\theta_2, \theta_1^5\theta_3, \theta_1^6\theta_3, \theta_1\theta_2^4, \\ \theta_1\theta_3^4, \theta_2^3\theta_3, \theta_2\theta_3^3, \theta_2^3\theta_3^4, \theta_2^4\theta_3^3, \theta_2^4\theta_3^2, \theta_2^4\theta_3, \theta_2^2\theta_3^4, \theta_2\theta_3^4, \theta_2^2\theta_3^2, \theta_1^4\theta_2^2, \theta_1^3\theta_2^2, \theta_1^4\theta_3^2, \theta_1^3\theta_3^2, \theta_1^2\theta_2^4, \theta_1^2\theta_3^4, \theta_1^2\theta_2^3, \theta_1^5\theta_2^2, \theta_1^5\theta_3^2, \\ \theta_1^4\theta_2^3, \theta_1^2\theta_3^3, \theta_2^3\theta_3^3, \theta_1^4\theta_3^3, \theta_1^3\theta_2^3, \theta_1^3\theta_3^3, \theta_1^3\theta_2\theta_3, \theta_1^2\theta_2\theta_3, \theta_1^4\theta_2\theta_3, \theta_1\theta_2^2\theta_3, \theta_1\theta_2^3\theta_3, \theta_1\theta_2^4\theta_3, \theta_1\theta_2\theta_3^4, \theta_1\theta_2\theta_3^3, \theta_1\theta_2^2\theta_3^2, \\ \theta_1^4\theta_2^2\theta_3, \theta_1^4\theta_2\theta_3^2, \theta_1^2\theta_2^3\theta_3, \theta_1\theta_2^3\theta_3^2, \theta_1\theta_2^2\theta_3^3, \theta_1^2\theta_2\theta_3^3, \theta_1^2\theta_2^4\theta_3, \theta_1^2\theta_2\theta_3^4, \theta_1^2\theta_2^2\theta_3^2, \theta_1\theta_2^2, \theta_1\theta_3^2, \theta_1\theta_2\theta_3, \theta_1^3\theta_2^2\theta_3^2, \theta_1^2\theta_2^3\theta_3^2, \theta_1^2\theta_2^2\theta_3^3, \\ \theta_1^5\theta_2\theta_3, \theta_1^3\theta_2^3\theta_3, \theta_1^3\theta_2\theta_3^3, \theta_1\theta_2^3\theta_3^3, \theta_2^5, \theta_2^6, \theta_1\theta_2^5, \theta_1\theta_2^6, \theta_1^3\theta_3^4, \theta_2^6\theta_3, \theta_2^5\theta_3, \theta_1^3\theta_2^4, \theta_1^2\theta_2^5, \theta_2^5\theta_3^2, \theta_1\theta_2^5\theta_3, \theta_3^5, \theta_3^6, \theta_1\theta_3^5, \theta_1\theta_3^6, \\ \theta_2\theta_3^6, \theta_2\theta_3^5, \theta_1^2\theta_3^5, \theta_2^2\theta_3^5, \theta_1\theta_2\theta_3^5 \end{array}\right) \tag{36}$$

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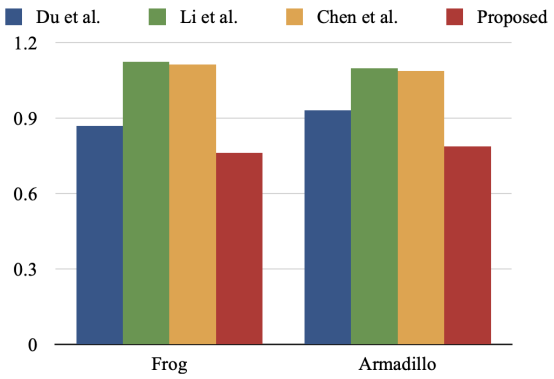


Fig. 5. Translation errors of the proposed method with representative methods. The unit is meters.

TABLE II
ROTATION AND TRANSLATION ERRORS FOR ANISOTROPIC REGISTRATIONS USING DIFFERENT METHODS (FROG/ARMADILLO)

Methods	Rotation Error (°)	Translation Error (m)
Du et al. [23]	1.567/1.808	0.869/0.932
Li et al. [24]	2.434/2.493	1.125/1.098
Chen et al. [25]	1.603/1.831	1.113/1.088
Proposed	1.218/1.226	0.763/0.791

the P3P [16], EPnP [17], Lambda-Twist P3P[3] [32] and our proposed solution (18) for comparisons. The intrinsic calibration is obtained using MATLAB calibration toolbox, i.e., the method of Zhang [33]. The corners of the pattern are extracted using the histogram method in [34]. Since the pattern is dark, direct corner extraction is trivial. We track the pattern using the fast correlation filter proposed in [35]. The trust region of the tracked area is used as the image for chessboard recognition and corner extraction (see Fig. 7). When implementing the P3P method, we use the random sample consensus (RANSAC) to select the best fourth point for the verified pose. After evaluation of each algorithm, the pose solution is refined by the same Quasi-Newton nonlinear optimizer by converting $\boldsymbol{q}$ to its Lie algebra $\boldsymbol{\xi}$. The algorithms are evaluated via the averaged loss function value

$$\mathcal{L}_{\text{PnP,mean}} = \mathcal{L}_{\text{PnP}}/\mathcal{N} \tag{43}$$

in (4). We use this metric because this is a mean quality value, which is proportional to the reprojection error. We name the proposed method as the algebraic PnP (APnP). The loss function values of various methods in a single experiment are shown, respectively, in Figs. 8–10.

The PnP reprojection errors are evaluated according to estimates given by various algorithms, whose snapshot is presented in Fig. 11. The root mean squared reprojection errors of various algorithms in pixels are presented in Table III. We use different markers to represent the reprojected corners from various algorithms. Among these reprojected markers, one may notice that some of them have a large bias with respect to the correct corners. The results indicate that the proposed APnP is very accurate and stable. The reason that

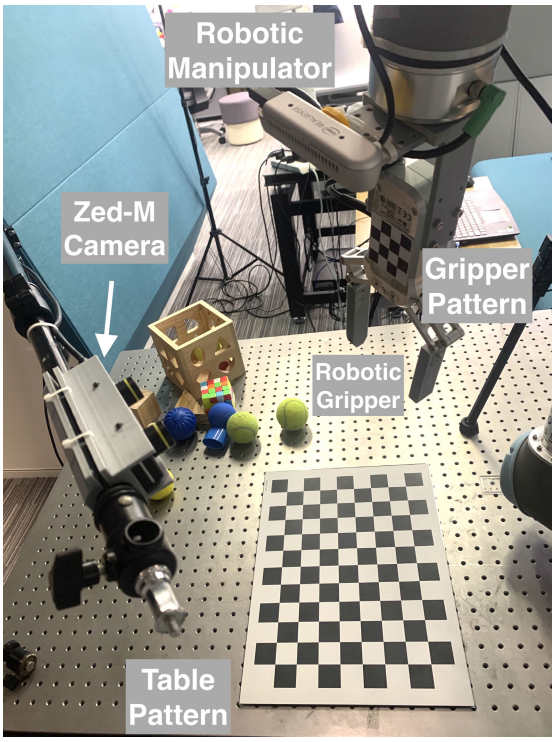


Fig. 6. Experimental platform for dynamic gripper pose tracking.

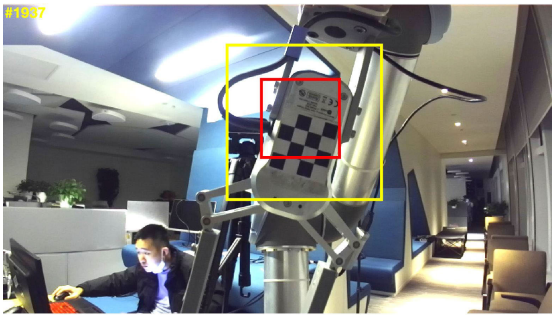


Fig. 7. Tracked chessboard pattern that attached to the robotic gripper. Red: tracked location of pattern; Yellow: Trust region of tracked location.

the accuracies of P3P and EPnP are low is that when the pattern moves, the extracted corner pixels have large errors due to possible motion blur. This can be visualized in the data provided in our open-sourced dataset (see Acknowledgment). The computational efficiency of the proposed APnP is high since it only requires EIG of a 4×4 matrix, which can be computed instantly via a fast method[4] reported in [14]. Therefore, the designed algorithm can be easily deployed to even low-configuration platforms for accurate camera pose estimation.

C. Application II: Magnetometer Calibration

For a vector magnetometer, the raw readings can be modeled as

$$\boldsymbol{m}^b = \boldsymbol{T}\boldsymbol{R}\boldsymbol{m}^r + \boldsymbol{b}_m + \boldsymbol{\epsilon}_m \tag{44}$$

[3]https://github.com/midjji/lambdatwist-p3p

[4]https://github.com/zarathustr/hand_eye_SO4/blob/master/eig4.m

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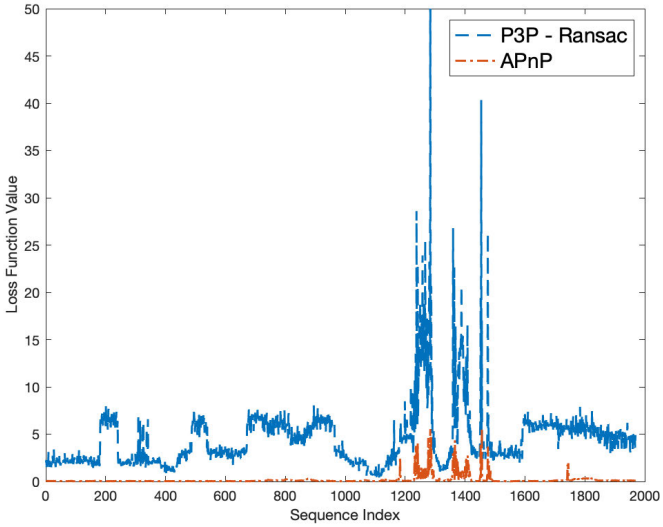


Fig. 8. Comparison of loss function values between P3P (with RANSAC) [16] and the proposed APnP.

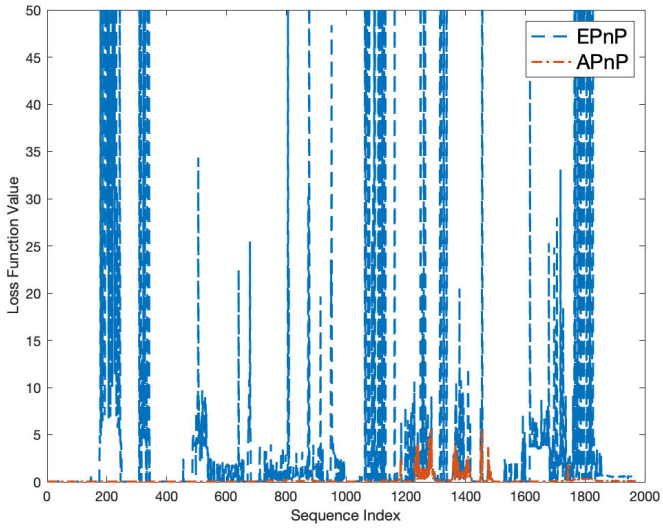


Fig. 9. Comparison of loss function values between EPnP [17] and the proposed APnP.

where $\boldsymbol{m}^b$ and $\boldsymbol{m}^r$ are 3-D vector measurements of magnetometer in the body frame and reference frame, respectively; $\boldsymbol{T} \in \mathbb{R}^{3\times3}$ stands for the calibration matrix that takes scale factor and nonorthogonality into account; $\boldsymbol{R}$ is the rotation matrix in SO(3); $\boldsymbol{b_m}$ and $\boldsymbol{\epsilon_m} \sim \mathfrak{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_{\boldsymbol{\epsilon_m}})$ denote the constant bias and stochastic noise term, respectively. The magnetometer calibration problem is to estimate unknown parameters $\boldsymbol{T}$, $\boldsymbol{R}$, $\boldsymbol{b_m}$ and $\boldsymbol{m}^r$, with given measurements of $\boldsymbol{m}^b$. The general calibration problem of the magnetometer can be parameterized as follows:

$$\underset{\substack{\boldsymbol{T}\in\mathbb{R}^{3\times3},\mathfrak{R}\in\mathrm{SO}^{\mathcal{N}}(3),\\ \boldsymbol{b_m}\in\mathbb{R}^3,\boldsymbol{m}^r\in\mathbb{R}^3}}{\arg\min} \quad \sum_{i=1}^{\mathcal{N}} \left\| \boldsymbol{m}_i^b - \boldsymbol{T}\boldsymbol{R}_i\boldsymbol{m}^r - \boldsymbol{b_m} \right\|^2 \tag{45}$$

in which $\mathfrak{R} \in \mathrm{SO}^{\mathcal{N}}(3) = \mathrm{SO}(3) \times \mathrm{SO}(3) \times \cdots \times \mathrm{SO}(3)$ stands for an element in a power manifold of SO(3) that includes all $\boldsymbol{R}_i$ for $i = 1, 2, \ldots, \mathcal{N}$ with order up to the measurement number $\mathcal{N}$, while i stands for the ith sensor sample time instant. Equation (45) is nonconvex, NP-hard, and usually trivial. As pointed out in [27], (45) suffers from nonunique solutions since $\boldsymbol{S}$ and $\boldsymbol{R}_i$ are coupled together. Special care has been taken to relax (45) to

$$\underset{\substack{\boldsymbol{U}\in\mathbb{T}(3),\\ \boldsymbol{b_m}\in\mathbb{R}^3,\tilde{\boldsymbol{m}}^n\in(\mathbb{U}^3)^{\mathcal{N}}}}{\arg\min} \quad \sum_{i=1}^{\mathcal{N}} \left\| \boldsymbol{m}_i^b - \boldsymbol{U}\tilde{\boldsymbol{m}}_i^n - \boldsymbol{b_m} \right\|^2 \tag{46}$$

where $(\mathbb{U}^3)^{\mathcal{N}}$ is the power manifold of unitary 3-D real vector space $\mathbb{U}^3$ with order $\mathcal{N}$ so that the ith element of $\boldsymbol{m}^n$ is $\boldsymbol{m}_i^n \in \mathbb{R}^3$, such that $\|\boldsymbol{m}_i^n\| = 1$; $\mathbb{T}(3)$ denotes the group of all real upper triangular matrices. In [27], it has been pointed out that, (46) can be interpreted into another relaxed optimization

$$\underset{\boldsymbol{\mathcal{U}}\in\mathbb{T}(3),\boldsymbol{b_m}\in\mathbb{R}^3}{\arg\min} \quad \sum_{i=1}^{\mathcal{N}} \left(1 - \left\|\boldsymbol{\mathcal{U}}\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right)\right\|^2\right)^2. \tag{47}$$

Detailed solutions to (47) have been given in [27], which achieve good accuracy for common datasets. Initial solution to (47) is given by $\boldsymbol{z} = [\mathrm{vec}(\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}})^\top, -2\boldsymbol{b_m}^\top\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}, \boldsymbol{b_m}^\top\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}\boldsymbol{b_m} - 1]^\top$, such that

$$\boldsymbol{Y}\boldsymbol{z} = \boldsymbol{0} \tag{48}$$

where $\boldsymbol{Y}$ is determined by measurements $\boldsymbol{m}_i^b$ for $i = 1, 2, \ldots, \mathcal{N}$. Solution of $\boldsymbol{z}$ can be sought by SVD of $\boldsymbol{Y}$ or EIG of $\boldsymbol{Y}^\top\boldsymbol{Y}$. However, when there are many outliers or insufficient measurements, the optimization can hardly be performed. From (48), it is able for us to see that since $\boldsymbol{z} \in \mathbb{R}^{10}$, at least ten noncoplanar measurements are required to obtain the initial solution. If the outlier rate is high or the measurements are not sufficient, the linear system (48) will become ill-posed, that is, $\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}$ in $\boldsymbol{z}$ cannot be guaranteed to be positive semidefinite. In this case, there is no such a Cholesky decomposition for a nonpositive semidefinite matrix. The origin of (47) is that, for ideal measurements, one has

$$\left\|\boldsymbol{\mathcal{U}}\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right)\right\| = 1. \tag{49}$$

Expanding (49), we have

$$\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right)^\top\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right) = 1. \tag{50}$$

Let $\boldsymbol{V}^\top\boldsymbol{D}\boldsymbol{V} = \boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}$ be an SO(3) EIG (SO(3)-EIG) of $\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}$ such that $\boldsymbol{D}$ denotes the matrix with diagonal entries of eigenvalues and $\boldsymbol{V} \in \mathrm{SO}(3)$ is the orthonormal basis of $\boldsymbol{\mathcal{U}}^\top\boldsymbol{\mathcal{U}}$. The SO(3)-EIG is not difficult since for one matrix $\boldsymbol{X}$ if $\boldsymbol{W}^\top\boldsymbol{D}\boldsymbol{W} = \boldsymbol{X}$, where $\det(\boldsymbol{W}) = -1$, $\boldsymbol{V}^\top\boldsymbol{D}\boldsymbol{V} = \boldsymbol{X}$ also holds for $\boldsymbol{V} = -\boldsymbol{W} \in \mathrm{SO}(3)$. Then, (50) denotes an ellipsoid such that the center is $\boldsymbol{b_v}$ and the semi-major axes are determined by the square roots of the diagonal elements in the inverse of $\boldsymbol{D}$. Equation (50) can be further explicitly given as

$$\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right)^\top\boldsymbol{V}^\top\boldsymbol{D}\boldsymbol{V}\left(\boldsymbol{m}_i^b - \boldsymbol{b_m}\right) = 1 \tag{51}$$

which can be treated as a unit sphere, say $\mathcal{X}$

$$\mathcal{X}\colon \boldsymbol{x}^\top\boldsymbol{x} = 1 \tag{52}$$

that is located in the origin by substitution of $\boldsymbol{x} = \pm\sqrt{\boldsymbol{D}}\boldsymbol{V}(\boldsymbol{m}_i^b - \boldsymbol{b_m})$. Therefore, (47) is actually an ellipsoid fitting problem and need not to be solved in the way of (48). For the model (51), we have a 9 degree-of-freedom for the unknowns, which is less than the number of unknowns in (48).

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TABLE III
ROOT MEAN SQUARED REPROJECTION ERRORS OF MULTIPLE METHODS (IN PIXELS)

Errors	P3P - RANSAC	EPnP	Lambda-Twist P3P	Proposed APnP
X-Axis	2.05076	0.22309	0.87190	**0.050636**
Y-Axis	2.17841	0.26945	0.86411	**0.052172**

The vector-field sensor intrinsic calibration problem turns into a surface registration problem with scale of $\mathcal{K} = \sqrt{\boldsymbol{D}} \in \text{diag}(\mathbb{R}^3_+)$, rotation of $\boldsymbol{R} = \boldsymbol{V} \in \text{SO}(3)$ and translation of $\boldsymbol{t} = -\mathcal{K}\boldsymbol{V}\boldsymbol{b_m} \in \mathbb{R}^3$, such that we find a rigid transformation that satisfies

$$\boldsymbol{x}_i = \mathcal{K}\boldsymbol{R}\boldsymbol{m}_i^b + \boldsymbol{t} \tag{53}$$

in which $\boldsymbol{v}_i^b$ on the ellipsoid surface is the ith corresponding point of $\boldsymbol{x}_i$ on the unit sphere (52), given by raw measurements $\boldsymbol{v}_i^b$ for $i = 1, 2, \ldots, \mathcal{N}$. Sometimes, when the three axes of the magnetometer are isotropic, $\mathcal{K}$ becomes a scalar, say s, such that $\boldsymbol{x}_i = s\boldsymbol{R}\boldsymbol{v}_i^b + \boldsymbol{t}$ and s actually represents the magnitude of the geomagnetic field at the local geodetic coordinates. The new problem is challenging, that is, it is now being formulated as a point-surface registration. We now solve it in a new geometrical manner. The point-surface registration is based on the fact that the geometry of the optimal surface formed by the points is homotopic to the surface to be registered. Note that here, homotopy is not rigorously identical to homeomorphism since the two registered surfaces do not always maintain the same inner volume. Therefore, we solve the problem in a discrete way. The following criteria have been proposed:

$$\underset{\substack{\boldsymbol{R}\in\text{SO}(3)\\ \boldsymbol{t}\in\mathbb{R}^3\\ \mathcal{K}\in\text{diag}(\mathbb{R}^3_+)}}{\arg\min} \quad \underset{\substack{\boldsymbol{r}_j\in\mathcal{X}\\ \boldsymbol{p}_j\in\tilde{\mathcal{Y}}\subset\mathcal{Y}}}{\arg\max} \sum \left\| \boldsymbol{r}_j - \mathcal{K}\boldsymbol{R}\boldsymbol{p}_j + \boldsymbol{t} \right\|^2. \tag{54}$$

The inner optimization maximizes the count of the points by selecting the most appropriate subset $\tilde{\mathcal{Y}}$ of the measurement set $\mathcal{Y}$, which forms an ellipsoidal surface that corresponds to the unit sphere $\mathcal{X}$. The outer optimizer then minimizes the objective function subject to the rigid loss defined in (53). In the inner loop, once a temporary $\tilde{\mathcal{Y}}$ has been found, the ellipsoid equation can be fit via (51) and thus a rough guess of s, $\boldsymbol{R}$, and $\boldsymbol{t}$ can be obtained. Then, via s, $\boldsymbol{R}$, and $\boldsymbol{t}$, the entire measurement set $\mathcal{Y}$ can be remapped to fit the unit sphere $\mathcal{X}$. In this way, by Cartesian distances of the remapped points to the unit surface, we can select the best points that are closest to $\mathcal{X}$. This results in a new map from $\mathcal{X}$ back to $\mathcal{Y}$, that recursively refines the two surfaces, which we name it for the first time, as the ICPs and surfaces (ICPSs). The kernel of this problem is solved via the proposed solver from (23) to (38).

To verify the proposed algorithm, we first conduct a hardware-in-the-loop (HITL) simulation. The 3DM-GX5-25 from Microstrain Inc., is employed to collect reference attitude information and calibrated magnetometer readings (see Fig. 12). The data is gathered at the frequency of 100 Hz. Then, the reference attitude matrices and magnetic data are utilized for the simulation of distorted magnetic readings, including anisotropic scale factor and bias. The distorted magnetometer data will be calibrated using the proposed method and the 3-D results are shown in Fig. 13. The magnetic norms before and after the calibration are shown in Fig. 14.

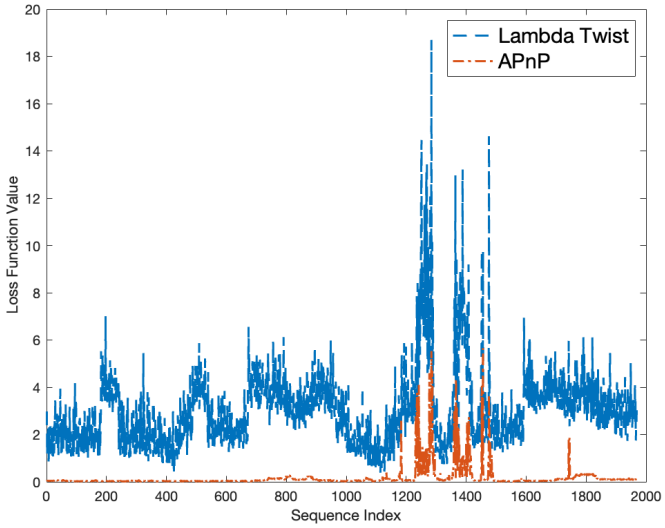


Fig. 10. Comparison of loss function values between Lambda-Twist P3P [32] and the proposed APnP.

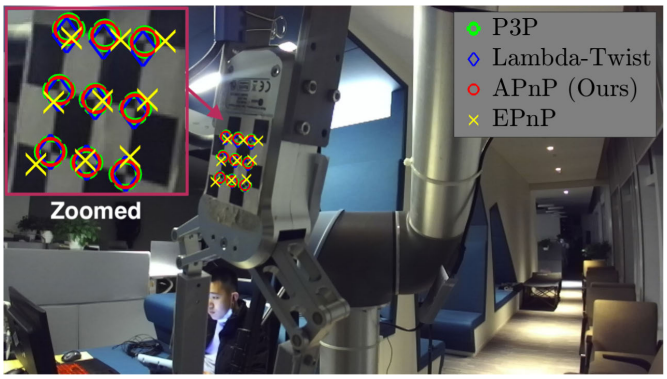


Fig. 11. Visualized reprojected markers using estimated poses from various algorithms.



Fig. 12. 3DM-GX5-25 IMU for HITL simulation.

We compare the proposed method with two previous methods by Wu and Shi [27] and Vasconcelos et al. [36]. After calibration, the calibration results from various methods are placed into a Kalman filter for attitude estimation from inertial and magnetic measurements [37]. During the HITL simulation, different outlier ratios are simulated. The results are shown in Table IV. The attitude errors are evaluated by the norm of the

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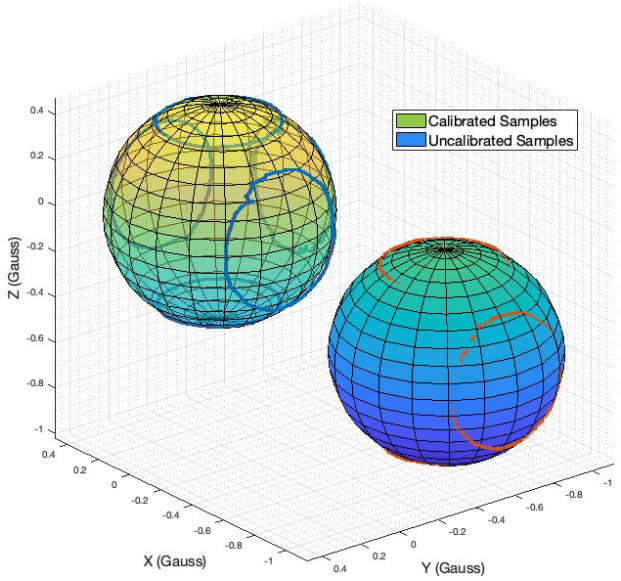


Fig. 13. Calibration results using the proposed method.

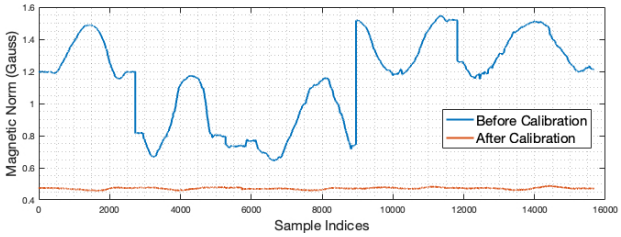


Fig. 14. Magnetic norm before/after calibration using the proposed method.

TABLE IV

ROOT MEAN SQUARED ATTITUDE ERRORS AFTER MAGNETOMETER CALIBRATION SUBJECT TO DIFFERENT OUTLIER RATIOS (IN DEGREE)

Outlier	Wu et al. [27]	Vasconcelos et al. [36]	Proposed
5%	0.54906	0.82433	**0.37805**
10%	2.34245	4.32126	**1.17829**
20%	6.57822	10.33268	**2.21061**

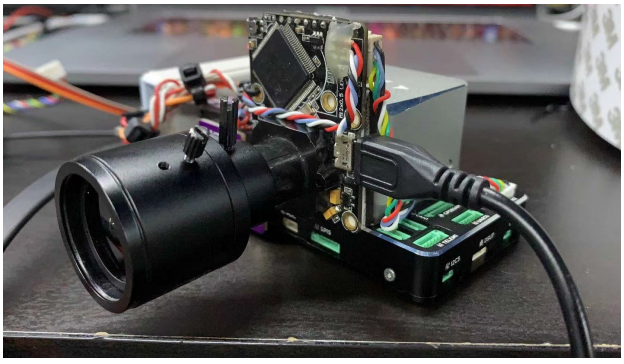


Fig. 15. Designed inertial /magnetic/visual odometry system.

angle axis (Rodrigues vector) of the estimated attitude matrix and reference one.

The results indicate that the proposed method outperforms these representative candidates. The main advantage is that the proposed method is able to estimate robust magnetometer calibration parameters subject to high outlier ratios, which indirectly verifies the effectiveness of the proposed solution to the anisotropic-scale registration problem.

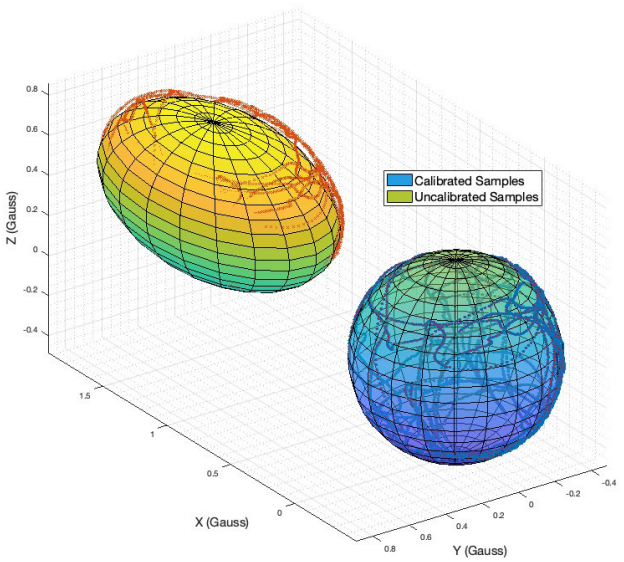


Fig. 16. Magnetometer calibration results of the designed odometry system.

TABLE V

TIME EFFICIENCY PROFILES OF VARIOUS ANISOTROPIC REGISTRATION METHODS (IN SECONDS)

Du et al. [23]	Li et al. [24]	Chen et al. [25]	Proposed
0.0923	0.0798	0.1247	0.0712

TABLE VI

TIME EFFICIENCY PROFILES OF VARIOUS PNP METHODS (IN SECONDS)

P3P - RANSAC [16]	EPnP [17]	Lambda-Twist P3P [32]	Proposed APnP
0.00285	0.00464	0.01832	0.00403

Moreover, the designed method is brought to our designed inertial/magnetic/visual odometry system (see Fig. 15). This system employs a magnetometer of RM3110, an inertial measurement unit (IMU) of ICM20948, and a global-shutter camera of MT9V034. The camera sensor needs much more power when capturing images than other inertial and magnetic sensors. This will cause sudden electromagnetic disturbances in the magnetic readings. Traditional methods like [27] and [36] cannot deal with the calibration problem very effectively for such a system. We use our proposed method to estimate the magnetic calibration parameters via (54). One of the many results will be presented in Fig. 16. Using the calibration results, we are able to compensate for the fast camera motion distortion by the magnetometer. The attitude estimation accuracy of the results with our calibration is 0.5203 degree on average for Euler angles and is 0.9811 degree with the method in [27], which verifies the superiority of our approach.

D. Time Efficiency

Time efficiency plays a vital role in PnPs, point cloud registration, and magnetometer calibration, as these processes often deal with vast amounts of data and require real-time performance. In PnPs, efficient algorithms for feature extraction, correspondence matching, and pose estimation are essential to ensure swift and accurate computation of camera pose from 2-D–3-D correspondences. Similarly, point cloud registration relies on optimized submodules for keypoint detection,

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TABLE VII
TIME EFFICIENCY PROFILES OF VARIOUS MAGNETOMETER CALIBRATION METHODS (IN SECONDS)

Wu et al. [27]	Vasconcelos et al. [36]	Proposed
0.6436	0.7689	0.9892

feature description, and transformation estimation to quickly align and fuse multiple point clouds into a coherent 3-D model. Magnetometer calibration also demands time-efficient techniques for data collection, noise filtering, and parameter estimation to promptly compensate for sensor errors and ensure precise orientation tracking. By prioritizing time efficiency and optimizing submodules, these processes can deliver high-quality results with minimal latency, enabling their effective deployment in various applications such as augmented reality, autonomous navigation, and geospatial mapping. Through previous experimental validations, we also test the time efficiency of various algorithms. Specifically, we summarize the results in Tables V–VII.

In each test, we test various candidates for 1000× to get the average time efficiency profiles. The time efficiency evaluation was conducted on a machine with an i7-8500 central processing unit, 16 GB RAM, and 1 TB SSD disk storage. Seen from the tables, it is found out that the proposed method has better time efficiency than other methods in 3-D registration and PnP tasks. However, it is notable that in magnetometer calibration tasks, the proposed method is slower than the existing representatives. This is because the proposed one requires more iterations to achieve good inlier estimation over large datasets while the kernel computation module, being anisotropic, indeed requires a higher computational burden than those analytical guesses in existing works. Therefore, there is a tradeoff between the time efficiency and accuracy of the sensor calibration tasks. For other tasks, the proposed method maintains faster than representatives.

V. CONCLUSION

After revisiting traditional similarity transformation problems, we prove that two representative branches, i.e., scale-SPCR and PnPs, can be solved in a unified manner. Furthermore, it is shown that an extended anisotropic-scale problem is challenging, and detailed globally optimal solutions are derived to solve this problem. The developed approach has been successfully applied to industrial robotic grasping tasks and magnetometer calibration. Future efforts will be paid to finding more potentially better solutions to these problems. Partial experimental data and codes of this article are open-source at `https://github.com/zarathustr/APnP`.

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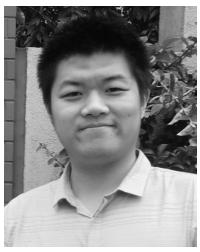
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